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FY IT

Assignment 1

Q1] Let the universal set be $U = \{1, 2, 3 \cdots 20\}$
Let A be the set of multiples of 2 in U,
B be the set of multiple of 3 in U, C be the set of multiples of 5 in U

a) List the element of A, B, C find $A \cap B \cap C$.
b) find $(A \cup B)'$ and $A' \cap B'$, hence verify that $(A \cup B)' = A' \cap B'$

→ Given : $U = \{1, 2, 3, 4, \cdots 20\}$
$A = \{2, 4, 6, 8 \cdots 20\}$
$B = \{3, 6, 9, 12, \cdots 18\}$
$C = \{5, 10, 15, 20\}$

To find : a] $A \cap B \cap C$
b] $(A \cup B)' = A' \cap B'$

Solution : a] $A = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$
$B = \{3, 6, 9, 12, 15, 18\}$
$C = \{5, 10, 15, 20\}$

$A \cap B = \{6, 12, 18\}$

$A \cap B \cap C = \{\emptyset\}$ $\emptyset$ or $\{\}$

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b] $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}$

$(A \cup B)' = U - (A \cup B)$

$= \{1, 2, 3 \cdots 20\} - \{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}$

$= \{1, 5, 7, 11, 13, 17, 19\}$

$A' = U - A$

$= \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$

$B' = U - B$

$= \{1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20\}$

$A' \cap B' = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\} \cap$
$\{1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 2$

$= \{1, 5, 7, 11, 13, 17, 19\}$

$\therefore (A \cup B)' = A' \cap B'$ ... Demorgan's law

Hence proved.

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Q2] set A has $|A| = 6$;

(a) find number of elements in the power set of A, the number proper subset of A number of subset of A containg 3 element.

(b) draw venn diagram shade regions representing $(P \cap Q) \cup (R - Q)$

→ Given :- $|A| = n(A) = 6$

To find :- • $n(P(A))$ • n(Subset of A) • no. subset of A with 3 elements

b] Shade region.

Solution :-

for $n(A) = 6$, number of subset of A $= 2^n$ where n is $n(A)$

∴ $n(A)$ No. of subset of $A = 2^6 = \underline{64}$

But proper subset means we dont count null set

∴ no. of proper subset of $A = 64 - 1 = \underline{63}$

$n(P(A)) = n$(Subset of A)

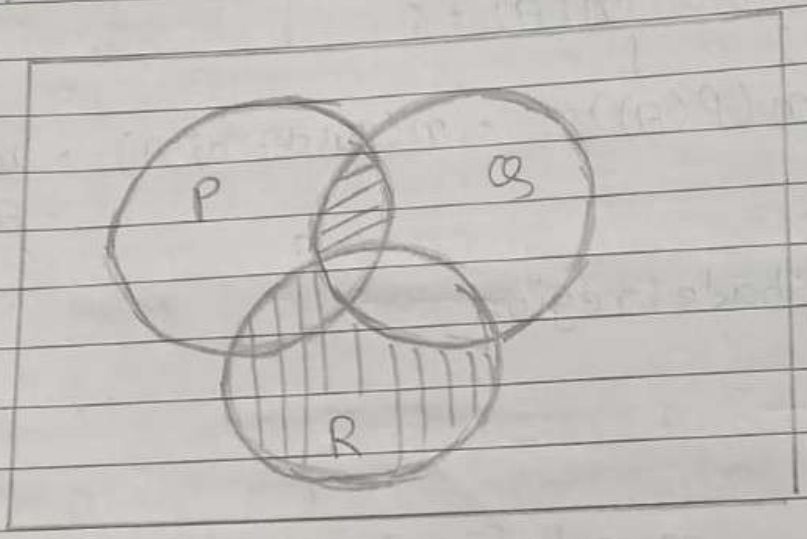
∴ $\underline{n(P(A)) = 64}$

number of subset containing 3 element $= {}^{6}P_3 = \frac{6 \times 5 \times 4}{1 \times 2 \times 3}$

$= \underline{20}$

Q3

(b) :-



$P \cap Q$:- ////

$R - Q$:- |||||

$(P \cap Q) \cup (R - Q)$:- the combined shaded area (region)

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Q3] Construct truth table for the statement $(P \vee q) \wedge (\sim P \vee r)$ and determine if it's logically equivalent to $q \vee (\sim P \wedge r)$.

→ P	q	r	~P	P∨q (1)	~P∨r (2)	(1)∧(2) [A]	~P∧r (3)	q∨(3) [B]
T	T	T	F	T	T	T	F	T
T	T	F	F	T	F	F	F	T
T	F	T	F	T	T	T	F	F
T	F	F	F	T	F	F	F	F
F	T	T	T	T	T	T	T	T
F	T	F	T	T	T	T	F	T
F	F	T	T	F	T	F	T	T
F	F	F	T	F	T	F	F	F

By looking at [A] and [B]

$(P \vee q) \wedge (\sim P \vee r)$ and $(q \vee (\sim P \wedge r))$

we can conclude that A and B are not logically equivalent.

Q4] Statement : "If a triangle is equilateral, then it is isosceles".

(a) write the converse, inverse and contrapositive and state truth value of each. given that original statement is True.

(b) express biconditional statement corresponding to : "A number n is even if and only if n^2 is even" state if it's true.

—> (a) Let p : A triangle is equilateral
Let q : A triangle is isosceles.

Then given statement : $p \rightarrow q$

(i) Converse $(q \rightarrow p)$:

• "if a triangle is isosceles, then it is equilateral".

• Truth value : false

(ii) inverse $(\sim p \rightarrow \sim q)$:

• "if a triangle is not equilateral, then it is not isosceles".

• Truth Value : false.

(iii) contrapositive ($\sim q \rightarrow \sim p$) :

- "If a triangle is not isosceles, then it is not equilateral"

- Truth value : True.

(b) - Expression : "A number n is even if and only if n^2 is even"

n is even $\longleftrightarrow$ n^2 is even.

- Truth value : True.

if n is even ($n = 2k$), then $n^2 = 4k^2 = 2(2k^2)$ which is even.

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Q.5] verify logical equivalence

(a) $\sim(P \wedge q) \equiv \sim P \vee \sim q$

(b) $\sim(P \vee q) \equiv \sim P \wedge \sim q$

—> (a) :

P	q	~P	~q	P ∧ q	~(P ∧ q) (1)	~P ∨ ~q (2)
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

from (1) and (2) we can verify that $\sim(P \wedge q)$ and $\sim P \vee \sim q$ are logically equivalent

$\therefore \sim(P \wedge q) \equiv \sim P \vee \sim q$

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(b) :

p	q	$\sim p$	$\sim q$	$p \vee q$	$\sim(p \vee q)$ (1)	$\sim p \wedge \sim q$ (2)
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

From (1) and (2) we can verify that

$\sim(p \vee q)$ and $(\sim p \wedge \sim q)$ are logically equivalent

$\therefore \sim(p \vee q) \equiv \sim p \wedge \sim q$

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Q6] A product code consists of 2 letters (26 english alphabet) followed by 3 digit (0-9)

(a) Total no. of product code (repetition allowed)
(b) Total no. of product code (repetion not allowed)
(c) no. of product digit are even.

→ Total available letter = 26
Total available digit = 10

Product code : _ _ _ _ _

letter digit.

(a) :- Since repetition is allowed total available letter and digit do not change.

∴ :- $26 \times 26 \times 10 \times 10 \times 10$
:- $26^2 \times 10^3$

:- 676,000

(b) :- repetition not allowed available no. of letter and digit will decrease gradually

:- $(26 \times 25)\ (10 \times 9 \times 8)$

:- 468,000

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(c) Repetition allowed but digit part is even.

letter part :- 26×26
:- 676

Digit part :- $10 \times 10 \times \frac{5}{or}$

for no. to be even its last digit should be even (0, 2, 4, 6, 8 for this case)

∴ Total no. of product code when repetition is allowed and digit part is even = 338,000

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Q7] (a) how many ways can the letters be arranged of the word "COMBINATORICS" so that all vowels come together.

→ Vowels in the word :- A, I, I, O, O
Combinatorics

group the 5 vowels as x, so now the total no. of letters are

Cmbntr$x$CS :- 9 letters.

in these 9 letter C is repeating twice

arrangement of these 9 letters will be $= \frac{9!}{2!} = 181,440$

But the vowels inside x will also be arranged in different ways.

arrangement of vowels in x $= \frac{5!}{2!\,2!} = 30$

as O and I are repeating twice.

now, Total arrangement $= 181440 \times 30$
$= 5,443,200$

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(b) number of non-negative integer solution of equation $x_1 + x_2 + x_3 + x_4 = 15$

→ formula for this exact stars-bars problem

$^{n+r-1}C_{r-1}$ where $n = 15$
$r = 4$

$\therefore {}^{15+4-1}C_{4-1} = {}^{18}C_3 = \frac{18 \times 17 \times 16}{1 \times 2 \times 3}$

$= 816$

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Q8] (a) coefficient of x^5y^7 in expansion $(2x-3y)^{12}$
(b) term independent of x in $(x^2-3/x)^9$

→ (a) :-

$T_{r+1} = {}^nC_r \, x^{n-r} \, y^r$

for our case, $n = 12$, & $r = 7$ from x^5y^7

$\therefore T_8 = {}^{12}C_7 \, x^5 \, y^7$

$\therefore$ coefficient of x^5y^7 is ${}^{12}C_7 =$

${}^{12}C_7 = 792$ for x^5 and y^7

But, for $2x$ and $-3y$ it will be

$792 \times 2^5 \times (-3)^7$

$\therefore$ coefficient of $x^5y^7 = -55,429,328$
in $(2x-3y)^{12}$

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(b) :-

$$T_{r+1} = {}^9C_r \, (x^2)^{9-r} \, (-3(x^{-1}))^r$$

$$= {}^9C_r \, x^{18-2r} \cdot (-3)^r \, x^{-r}$$

$$T_{r+1} = {}^9C_r \, (-3)^r \cdot x^{18-3r}$$

for x to be 1; $18 - 3r = 0$

$\therefore r = 6$

$\therefore T_7 = {}^9C_6 \, (-3)^6$

$= 84 \cdot 729$

$= 61,236$

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Q9] $S = \{1, 2, 3 \cdots 20\}$

(a) :- prove pigeonhole principle, 11 integer are seleted from S, at least two of the selected integers must add up to 21.

(b) :- using pigeonhole principle, prove any 13 people at least 2 have been born in same calendar month

→ (a) :- group of 20 such that $x + y = 21$

$\{1, 20\}$ $\{2, 19\}$ $\{3, 18\}$, $\{4, 17\}$ $\{5, 16\}$ $\{6, 15\}$

$\{7, 14\}$ $\{8, 13\}$ $\{9, 12\}$ $\{10, 11\}$ ~~$\{11$~~.

There are exactly 10 pairs (pigeonholes)

The selected integer represent pigeons.

By pigeonhole principle, when you distribute 11 pigeons into 10 pigeonholes (pairs) at least one pair must contain 2 selected integers who adds upto 21.

(b) :- 12 Calendar months in year (pigeonholes)
13 people (pigeons)

Since the number of pigeons (13) is strictly greater then number of pigeonholes (12),
The pigeonhole principle gurantees that atleast one month must contain more than one person or atleast 2 person must share the same birth month.

Q10] A
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→

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Q10] A bag contains 5 red, 4 blue, 6 green balls Two balls are drawn at random from bag without replacement.

(a) find the probability that both balls drawn red
(b) Probability that 2 balls are of different colors
(c) Probability of complement of the event "at least one ball drawn is green".

→ Red ball :- 5
Blue ball :- 4
Green ball :- 6
} Total balls :- 15

Total outcomes :- $^{15}P_2 = \frac{15 \times 14}{1 \times 2} = 105$

(a) :- favourable ways :- Both are red

$P(A) :- \underbrace{\frac{5}{15} \times \frac{4}{14}} = \frac{2}{21}$

∵ no replacement initial 5 favourable and 15 Total will be reduced to 4 favourable and 14 total

(b) :- P(2 diff balls) = 1 - P(same color balls)

P(same color balls) = P(same red) + P(both B) + P(both G)

$= \frac{2}{21} + \frac{4}{15} \times \frac{3}{14} + \frac{6}{15} \times \frac{5}{14}$

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$$\frac{2}{21} + \left(\frac{2}{35}\right) + \frac{1}{7}$$

$$\approx 0.7 \quad 0.28$$

$$\therefore [P(\text{Diff color})] = 1 - P(\text{same color})$$

$$\approx 1 - 0.28$$

$$\approx 0.72$$

(c) :- Total balls = 15

non-green balls = 9

(as complement of atleast one green is no green)

$$P(\text{1st ball}) = \frac{9}{15}$$

$$P(\text{2nd ball}) = \frac{8}{14}$$

$$P(\text{no green}) = \frac{9}{15} \times \frac{8}{14} = \frac{12}{35}.$$